Gonadal PIP-seq reveals genes enriched in germ cell development and sexual differentiation in sea lamprey (*Petromyzon marinus*)

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Abstract:

Vertebrate sex determination is highly diverse, with distinct mechanisms that have evolved repeatedly and independently across phylogeny. The sea lamprey (*Petromyzon marinus*) is an extant jawless vertebrate that remains sexually labile during its larval stage. Their gonad contains bipotential, female, and male germ cells before metamorphosis. To examine the genetic programs

in germ cells at different developmental stages, we profiled transcriptomes of dissociated gonadal cells from larvae and recently metamorphosed juvenile transformers (sex-determined) using fluorescence-activated cell sorting (BD FACS Aria IIu) and PIPseq^{TM} (Fluent BioSciences/Illumina). After removing the doublets, samples were merged and 60,866 cells were clustered into 18 cell clusters according to their gene expression profiles. Primordial germ cells (PGCs) exhibited enriched expressions of genes involved in cell fate commitment, cannotical Wnt signaling pathway, iron ion homeostasis, regulation of RNA splicing and mRNA processing, and protein folding and catabolic processes. These factors provide possible interactions between environmental factors, cellular functions, and cell fate determinants during PGC development and differentiation. Germ cells in larvae had enriched expression of genes involved in DNA replication, transcription, translation, cell cycle, cell differentiation, cell adhesion and migration, and chromatin remodeling. Specifically, female germ cells differentially expressed *aep1*, *cuzd1*, *s100a1*, and *zp* genes from multiple loci, whereas male germ cells differentially expressed *piwil1*. Pseudotime trajectory and Gene Ontology (GO) analyses revealed that genes with changing expression along the female differentiation pathway were involved in binding of sperm to zona pellucida, egg coat formation, prevention of polyspermy and positive regulation of acrosome reaction. On the other hand, genes with changing expression along the male differentiation pathway were involved in cell cycle, DNA replication and repair, transcription and mRNA splicing and rRNA maturation, ribosome assembly and translation, and protein folding, modification and catabolism. Our results suggest that sea lamprey sex determination and sexual differentiation likely involve the coordinated action of numerous genes that regulate germ cell specification and development.

Keywords: single-cell RNA-seq, Wnt signaling, primordial germ cell, oogonia, spermatogonia, oocyte, spermatocyte

Introduction

The mechanisms of sex determination and sexual differentiation are strikingly variable among animals. Sex determination systems are typically classified as genetic sex determination (GSD) or environmental sex determination (ESD). However, many vertebrate species use both GSD and ESD mechanisms simultaneously (Ezaz, 2006; Holleley, 2015). In vertebrates, different cell types in the gonad can initiate the process of sex determination. In mammals, somatic cells seem to induce sexual differentiation processes, whereas in fishes germ cells appear to drive the processes of sexual differentiation (Capel, 2017). During sexual differentiation, the reproductive system contains bipotential primordial germ cells (PGCs). As a result, each embryo has the potential to differentiate into either male or female (Capel, 2017). In gonochoristic species, sex determination gene or genes activate sexual differentiation pathway of one sex and inhibit that of the opposite sex. In sea lamprey, a jawless vertebrate residing at the earliest branching point of vertebrate phylogenetic tree (Delsuc et al., 2006), female and male differentiation pathways may coexist during early gonadal development, but the male pathway seems to require more germline-specific genes to initiate or maintain the male sexual differentiation, or both (Timoshevskiy et al., 2025; Yasmin et al., 2022). Therefore, sea lamprey offers a valuable opportunity to investigate the evolution of vertebrate sex determination mechanisms.

Lampreys are characterized by a long larval period with undifferentiated gonads (Okkelberg 1921; Hardisty 1965). After metamorphosis, lamprey commit to a specific sex, and the gonad differentiate into either the ovary or testis (Lowartz & Beamish, 2000). Parasitic lamprey species

usually develop a large number of oocytes in the gonad compared to the non-parasitic lamprey species (Beamish & Potter, 1975). Although the underlying sex determination mechanisms in lampreys are unclear, environmental factors such as population density and growth rate seem to bias the sex ratio in the population (Johnson et al., 2017). Under conditions promoting rapid growth, the percentage of males varied directly with density and inversely with temperature (Docker & Beamish, 1994). However, temperature had little effect when larval growth was slow (Beamish, 1993).

To uncover the gene expression pathways associated with primordial, male and female germ cells in sea lamprey, we used Particle-templated Instant Partition sequencing (PIP-seq; Clark et al., 2023), a microfluidics-free single-cell RNA-sequencing technique to examine dissociated gonadal cells from sex-undetermined larvae and sex-committed female and male transformers. We identified gene expression profiles and the genetic program and pathways specific for female vs. male germ cells. The expression pattern of selected genes in sea lamprey gonad was confirmed using immunohistochemistry and immunofluorescence staining, and Hybridization Chain Reaction fluorescence *in situ* hybridization (HCR-FISH). This study provides valuable resources to investigate sex determination and sexual differentiation in the sea lamprey, a notorious invasive species in the Great Lakes, and may aid the long-term population control effort instigated jointly by the United States and Canada.

Methods and Materials

Animals:

Larval (sex-labile; total length: 11.2 ± 0.1 cm, body weight: 2.0 ± 0.0 g; n = 12, collected in July, 2023) and newly transformed sea lamprey (sex-determined, referred as transformers; total

length 13.1 ± 0.1 cm, body weight: 1.9 ± 0.1 g; n = 12, collected in November, 2023) for PIP-seq and additional animals in different developmental stages (larvae, total length: 9.2 ± 0.1 cm, body weight 1.2 ± 0.0 g, n = 60; transformers, total length: 12.6 ± 0.3 cm, body weight: 5.9 ± 0.1 g, n = 20; adult sea lamprey, total length: 50.3 ± 2.0 cm, body weight: 211.0 ± 38.0 g, n = 10; collected in 2022-2023) were provided by the staff of the U.S. Geological Survey, Hammond Bay Biological Station, Great Lakes Science Center, Millersburg, MI, U.S.A. Animals were transferred to Michigan State University where standard operating procedures for transporting, maintaining, handling, anesthetizing, and euthanizing sea lamprey were approved by the Institutional Committee on Animal Use and Care of Michigan State University and in compliance with standards defined by the National Institutes of Health Guide for the Care and Use of Laboratory Animals (Institute for Laboratory Animal Research, 2011). All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

Gonadal cell dissociation:

Six transformers of each sex and 12 larvae were euthanized with 0.65 mg/ml ethyl 3-aminobenzoate methanesulfonate (MS-222, Sigma-Aldrich Co. LLC, St. Louis, MO, U.S.A.). Gonadal tissues were dissected out and placed in a 5 ml propylene tube containing 3 ml HBSS (ThermoFisher; Life Technologies Corporation, Carlsbad, CA, U.S.A.) on ice (3 gonads/tube). Gonadal cells were dissociated using a ThermoScientific™ Pierce™ Primary Neuron Isolation Kit (ThermoFisher No. 88280) according to the manufacturer's instructions. Briefly, gonadal tissues were rinsed with 4 ml HBSS, then incubated in 1 ml neuronal isolation enzyme with papain reconstituted in HBSS in an incubator shaker (15 rpm) at room temperature for 30 min.

Papain works the best compared to other enzyme mixtures tested in the preliminary dissociation experiments, including Miltenyi Multi Tissue Dissociation Kit 3, and various mixtures of collagenases, dispase, trypsin, and elastase. Dissociated gonadal cells were centrifuged at 200 x g, 4 °C for 3 min. The supernatant was decanted, and 1 ml HBSS was added to the pellet and centrifuged at 200 x g, 4 °C for 3 min. The resulting supernatant was decanted, and 500 µl of serum supplemented Neuronal Culture Media (10 % fetal bovine serum, 1 x glutamine supplement, and 1 % penicillin/streptomycin) was added to the tissue pellet. The enzyme-digested gonadal tissues were broken down by pipetting up and down 20 times using a 1000 µl wide-bore pipette tip set at 400 µl. After that, 500 µl of serum supplemented Neuronal Culture Media was added to the suspension, and the suspension was pressed through a 40 µm BelArt™ Flowmi™ tip strainer (ThermoFisher). Cell concentration and cell viability were examined using a BioRad TC20 Automated Cell Counter (BioRad Laboratories, Corp., Hercules, CA, U.S.A.). Cell suspension was centrifuged at 200 x g, 4 °C for 3 min. The supernatant was discarded, and flow cytometry cell sorting buffer (HBSS with 1% heat-inactivated fetal bovine serum, 1 mM EDTA, and 25 mM HEPES, pH 7.0) was added to the pellet to adjust the cell concentration to around 1 x 10^6 cells/ml and split into several 5 ml propylene tubes. One tube served as the unstained control, and the other tube was added with 2 µl 7-aminoactinomycin D (7-AAD, ThermoFisher)/ml cell suspension for Fluorescence-activated Cell Sorting (FACS) using a BD FACS Aria IIu (BD Biosciences, San Jose, CA, U.S.A.). Live cells (7-AAD negative) were collected in a 5 ml propylene tube coated with 500 µl fetal bovine serum.

PIP-seq:

Samples were prepared using the PIPseq™ T20 3′ Single Cell Capture and Lysis Kit v4.0 following the instructions provided by Fluent Biosciences (now part of Illumina, Inc., San Diego, CA, U.S.A.). Briefly, FACS-collected cells were centrifuged at 200 x g, 4 °C for 3 min. The supernatant was discarded, and 1 ml Cell Suspension Buffer was added and mixed 5 times using a P1000 wide-bore low retention pipette tip. The cell suspension was centrifuged at 200 x g, 4 °C for 3 min, and the supernatant was discarded. 400 µl Cell Suspension Buffer was added to the pellet and mixed 15 times using a standard-bore P1000 pipette tip to create a homogeneous cell suspension. The cell suspension was pressed through a 40 µm Flowmi™ tip strainer. Cell concentration and viability was determined using a BioRad TC20 automated cell counter. Cell concentration was adjusted to 4 x 10^6 cells/ml, and 10 µl of the cell suspension was added to the PIPs and mixed 10 times with a P200 pipette set to 180 µl. 1000 µl partitioning reagent was added to the cell/PIP mixture and vortex horizontally at 2000 rpm for 15 sec and vertically at 2000 rpm for 2 min using the Starter kit equipment (Fluent BioSciences/Illumina). The PIP tubes were sat on a provided stand for 30 seconds, and extra partitioning reagents were removed using a provided syringe with G22 blunt tip needle. 630 µl chemical lysis buffer emulsion was added to the top of the PIP emulsion, and the emulsion was incubated in the dry bath provided in the Starter kit using Program A and shipped to the company for further processing (library preparation, sequencing, and data analysis). Four samples were submitted: C00003262 (larval technical duplicate 1, submitted on July 11, 2023), C00004336 (larval technical duplicate 2, submitted on July 11, 2023), C00004337 (female transformer, submitted on November 8, 2023), and C00004374 (male transformer, submitted on November 8, 2023). Data were deposited in

National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) (https://www.ncbi.nlm.nih.gov/geo/) under GSE334273.

Data analysis:

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PIPseeker 3.0.5_Mk6.1 (https://www.fluentbio.com/products/pipseeker-software-for-data-analysis/) was used to quantify raw reads against kPetMar1.pri (GCF_010993605.1). The following analyses were performed using R Statistical Software (v4.4.1) (R Core Team, 2021). The code for all analyses are available on GitHub at https://github.com/MSUBioinformaticsCore/BCC125_sea_lamprey_scRNA. Supplementary tables and figures are available on Dryad Repository at DOI: 10.5061/dryad.c59zw3rrc.

Quality control

Quality control steps were performed for each sample separately. To differentiate genuine cells from empty droplets, the emptyDrops function from the *DropletUtils* (v1.26.0) package was used, setting the "by.rank" parameter to 20K cells (Lun et al., 2019). Barcodes with an FDR < 0.001 were retained as likely cell-containing droplets. Quality control metrics were computed using perCellQCMetrics from the *scater* (v1.34.1) package (McCarthy et al., 2017). Mitochondrial genes were identified as those on the mitochondrial chromosome (NC_001626.1). Cells with percent of mitochondrial transcripts > 3 median absolute deviations (MAD) above the median percent mitochondrial transcripts, as calculated by the isOutlier function from the *scater* package, were removed (McCarthy et al., 2017).

Normalization and clustering

Normalization steps were performed using the *scran* (v1.34.0) package (Lun et al., 2016). Counts from each sample underwent log-normalization using logNormCounts. Feature selection was performed on each dataset individually by modeling gene variances with modelGeneVar, followed by selecting the top 10 % highly variable genes (HVGs) via getTopHVGs. To merge individual datasets and correct for batch effects, a common set of genes was identified across all samples via intersection. Using this gene universe, expression matrices and variance statistics for each sample were subset to include only shared genes. To account for depth differences, we applied multiBatchNorm from the *batchelor* (v1.22.0) package (Haghverdi et al., 2018) across the samples. HVGs were then re-identified using the combined variance model generated via combineVar from the *scran* package (v1.34.0), and the top 5,000 HVGs were retained (Lun et al., 2016). Sample integration was conducted using the fastMNN method from the *batchelor* (v1.22.0) package, using the top HVGs as input features (Haghverdi et al., 2018). The resulting merged dataset, containing batch-corrected PCA embeddings, was clustered using a shared nearest neighbor graph (k = 30) followed by Louvain clustering.

Identification of cluster enriched genes

To identify enriched genes for each cell cluster in the integrated dataset, we performed differential expression (DE) analysis using the findMarkers function from the *scran* (v1.34.0) package using each sample's normalized log-expression values (Lun et al., 2016). A two-sample t-test was used for each gene to pairwise compare expression in each cluster against all others. To account for inter-sample variability, we applied blocking by sample origin. The direction="up" parameter was used to focus on genes that were upregulated in each cluster

compared to others. The pval.type="some" option was used to retain genes with statistically significant differential expression in some but not necessarily all pairwise comparisons, enhancing sensitivity in heterogeneous datasets. Clusters were annotated by comparing cluster enriched genes with cell-type marker genes collected from the literature (Supplementary Table S1).

Gene Ontology (GO) Annotation

To assist in annotating cell type clusters, we performed GO enrichment analysis on the top 100 marker genes for each cluster using a hypergeometric test for overlap with the total number of assayed genes as the gene universe (21,967). GO annotations for *Petromyzon marinus* genes were retrieved from https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/010/993/605/GCF_010993605.1_kPetMar1.pri/GCF_010993605.1_kPetMar1.pri_gene_ontology.gaf.gz. The annotations were preprocessed by filtering out entries with the "NOT" qualifier to remove explicitly negated annotations. To reduce noise and avoid overly broad or overly narrow terms, we filtered GO terms to include only those associated with more than 10 and fewer than 200 genes. False discovery rates (FDR) were calculated using the Benjamini-Hochberg correction. Only GO terms with p-value < 0.01 were retained for downstream analysis and cluster annotation.

Germ cell subclustering

Cells from the following germ cell clusters were selected from the full data set: cluster 5 (oogonia/oocytes), cluster 15 (mitotic primordial germ cells), cluster 16 (spermatocytes I), cluster 17 (spermatocytes II), and cluster 18 (migrating male germ cells). The raw counts were

renormalized and subclustered using the same methods as the full dataset. Briefly, normalization and variance modeling were carried out with *scran* v1.34.0 (Lun et al., 2016), and batch correction and integration were performed using *batchelor* v1.22.0 via fastMNN (Haghverdi et al., 2018). Clustering was based on a shared nearest neighbor graph (k = 8) and Louvain algorithm. Cluster-enriched genes were identified with findMarkers (*scran*), blocking by sample (Lun et al., 2016).

Trajectory analysis

We conducted a pseudotime analysis on integrated germ-cell RNA-seq data using the *TSCAN* (v1.44.0) package framework (Ji & Ji, 2017). Pseudotime was inferred using quickPseudotime applied to the integrated dataset. Specifically, a minimum spanning tree was constructed based on cluster centroids computed in the corrected dimensionality-reduced space from MNN integration. Pseudotemporal ordering was assigned to cells starting from a specified root node (PGCs in this case). We computed a consensus pseudotime ordering for visualization using averagePseudotime. Statistical testing of gene expression along each branch of the pseudotime trajectory was performed with the testPseudotime function using each sample's normalized log-expression values. GO enrichment analyses were performed as described above on the top 100 up- and down-regulated genes by FDR along each path.

LOC gene functional annotation

Human homologs or homologs from other species derived by Wang et al. (2024) were used to further annotate genes with LOC numbers in the NCBI sea lamprey reference genome

(kPetMar1.pri) when possible. See Supplementary Table S2 for a map from kPetMar1 gene IDs to Wang et al. (2024) gene IDs.

Immunohistochemistry and Immunofluorescent Staining

Sea lamprey gonadal tissues were fixed in 4% paraformaldehyde in 0.1M phosphate buffer (pH 7.4) for at least 24 h, rinsed with 0.1M phosphate buffer saline (PBS, pH 7.4) for 5 min, dehydrated in increasing ethanol series (10%~70% ethanol with 10% increment each for 5 min), and sent to Michigan State University Investigative Histopathology Laboratory to process for paraffin sections. Paraffin sections (on the slides) were deparaffinized in xylene for 5 min 3 times, rehydrated through 100% ethanol for 5 min, 95% ethanol for 5 min, 70% ethanol for 5 min, and PBS for 5 min twice.

Immunohistochemistry staining for selected proteins (from Supplementary Table S1) was conducted according to the manufacturer's instruction. Negative control with the normal goat serum as the blocking and primary antibody was stained simultaneously. Briefly, gonadal sections were incubated with BLOXALL Endogenous Blocking Solution (Vector Laboratories, Inc., Newark, CA, U.S.A.) for 10 min and washed in PBS for 5 min twice to eliminate endogenous peroxidases and phosphatases. Gonadal sections were then incubated in Avidin solution (Avidin-Biotin Blocking Kit, Vector) for 15 min, washed with PBS for 5 min, and incubated in Biotin solution (Avidin-Biotin Blocking Kit, Vector) for 15 min to eliminate endogenous biotin. After PBS wash for 5 min, the sections were incubated in primary antibody with normal goat serum (VECTASTAIN Elite ABC-HRP Kit, Vector) in PBS with 0.1% Triton X-100 (Sigma) at 4 °C overnight. The sections were then washed with PBS for 5 min 3 times, incubated in biotinylated goat-anti-rabbit secondary antibody with normal goat serum

(VECTASTAIN Elite ABC-HRP Kit, Vector) in PBS with 0.1% Triton X-100 for 30 min at room temperature, and washed in PBS for 5 min 3 times. The sections were incubated in ABC solution (VECTASTAIN Elite ABC-HRP Kit, Vector) for 30 min, washed with PBS for 5 min 3 times, subsequently incubated in ImmPACT DAB (ImmPACT DAB Peroxidase Substrate kit, Vector) for 2-10 min, washed in tap water for 5 min 2 times, and counter stained with Hematoxylin QS (Vector) for 45 sec. Gondal sections were washed with tap water, dehydrate through 70%, 95%, and 100% ethanol, 5 min each. The stained sections were clarified in xylene for 5 min 3 times, coverslipped with DPX mounting media (Sigma), and imaged using an ECHO Revolution imaging system (ECHO, a BICO Company, San Diego, CA, U.S.A.). Primary antibodies: 1:500 rabbit-anti-Cxcr4 (SAB3500383, Sigma); 1:1000 rabbit-anti-Rbpms2 (SAB2106506, Sigma); 1:1000 rabbit-anti-S100a1 (HPA006462, Sigma).

For immunofluorescent staining of selected proteins (from Supplementary Table S1), paraffin sections were deparaffinized, rehydrated, and incubated in primary antibodies as described above, and then incubated in fluorescent secondary antibody, and coverslipped using ProLong Diamond antifade mounting media with DAPI (ThermoFisher). Negative control slides were stained simultaneously with the normal goat serum as the blocking and primary antibody. Primary antibodies: 0.85 µg/ml rabbit-anti-Piwil1 (Invitrogen PA5-17034, ThermoFisher); 10 µg/ml rabbit-anti-Sycp3 (Bio-techne/Asuragen, Inc., Austin, TX, U.S.A.); 0.5 µg/ml rabbit-anti-Wnt5b (Invitrogen PA5-33114, ThermoFisher). Secondary antibodies: 1 µg/ml goat-anti-rabbit IgG-Alexa 594 (ThermoFisher).

HCR™GoldRNA-FISH (HCR)

Selected marker gene (from Supplementary Table S1) mRNA was detected using designed probes from Molecular Instruments, Inc. (Los Angeles, CA, U.S.A.) using mRNA sequence retrieved from the NCBI sea lamprey genome (GCF_010993605.1). The hybridization procedure followed the manufacturer's instruction. Negative control slides were stained simultaneously without adding the probes. Briefly, paraffin sections were prepared as described in Immunohistochemistry and Immunofluorescent Staining section. Wash buffer used PBS containing 0.1% diethylpyrocarbonate (DEPC) and 0.3% Tween 20 (PBST-DEPC). Gonadal sections were treated with 5 µg/ml proteinase K in PBST-DEPC at room temperature for 10 min, washed with PBST-DEPC for 5 min 3 times. Gonadal sections were prehybridized in a humidifying chamber with HCR™ HiFi Probe Hybridization Buffer at 37 °C for 10 min and hybridized with the designed probe mix (detail see below) in HCR™ HiFi Probe Hybridization Buffer in a humidifying chamber at 37 °C overnight. Gondal sections were washed with HCR™ HiFi Probe Wash Buffer at 37 °C for 15 min 4 times, incubated in HCR™GoldAmplifier Buffer at room temperature for 30 min in a humidified chamber, and continued incubation with fluorescent hair pin amplifier solution at room temperature overnight in the dark. Gonadal sections were washed with HCR™GoldAmplifier Wash Buffer at room temperature for 15 min 4 times, 5 x SSC at room temperature for 15 min 4 times, PBS for 15 min, coverslipped with ProLong Diamond antifade mounting media with DAPI (ThermoFisher) and imaged using a Leica Stellaris 8 DIVE imaging system (Leica Microsystems, Inc., Deerfield, IL, U.S.A.). Probe: *piwil1*-X8_Alexa 594.

Results and Discussion

PIP-seq identified germ and somatic cell types in sex-labile and sex-determined sea lamprey gonads

After doublet removal and quality filtering, PIP-seq captured 15,574 cells from C00004336 (technical duplicate 1 of pooled cells from 12 larvae), 15,840 cells from C00004362 (technical duplicate 2 of pooled cells from 12 larvae), 12,803 cells from C00004337 (pooled cells from 6 female transformers), and 16,649 cells from C00004374 (pooled cells from 6 male transformers) (Supplementary Table S3). The four datasets were merged, yielding 18 cell clusters (Figure 1). We assigned cluster identities using the expression of established cell-type markers (Supplementary Table S1), gene transcripts enriched within each cluster (Supplementary Table S4: cluster_marker_genes.csv), and GO terms overrepresented among those enriched genes (Supplementary Table S5: gene ontology csv file?). Interestingly, multiple *cuzd1* loci were expressed in cluster 5 (oogonia/oocytes), but one specific transcript (*cuzd1.15: LOC116956112*) was uniquely enriched in this cluster (Supplementary Table S4: cluster_marker_genes.csv).

To validate these assignments, we examined selected markers at both the protein level (immunohistochemistry and immunofluorescent staining) and the transcript level (HCR™ GoldRNA-FISH) in sea lamprey gonadal tissues, and in each case the staining patterns were consistent with the cluster identities. Germ cell marker *piwil1* was enriched in cluster 15 (mitotic primordial germ cells), and Piwil1-immunoreactivity was observed in migrating primordial germ cells and within the germ cell cyst (Figure 2 & Supplementary Figure S6). HCR-FISH for *piwil1* yielded concordant results (Figure 3). Likewise, *sycp3* was enriched in clusters 15, 17, and 18, and Sycp3-immunoreactivity was found in migrating PGCs, within the germ cell cyst, and in spermatocytes (Figure 4). Immunohistochemistry for Cxcr4, Rbpms, and S100a1 also confirmed

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the PIP-seq results (Supplementary Figures S7). Our data support the long-acknowledged notion that larval gonad contained bipotential germ cells (Okkelberg 1921; Hardisty 1965).

Pseudotime analysis identified divergent female and male germ cell trajectories

We then isolated data from germ cells (clusters 5: oogonia/oocytes; 15: mitotic primorodial germ cells; 16: spermatocytes I; 17: spermatocytes II; 18: migrating male germ cells; Figure 1) for further analysis (Figure 5; Supplementary Table S8: node_marker_genes.csv; Supplementary Table S9: node_marker_GO_pvalues.csv). Genes with expression changing along the developmental paths from PGCs to mature germ cells were identified using pseudotime trajectory analysis (Figure 6; Supplementary Table S9: pseudotime_marker_genes.csv; Supplementary Table S9: pseudotime_marker_GO_pvalues.csv). The top 25 genes exhibiting dynamic expression along the female pathway (ending in node 5.4; Figure 7) and along the male pathway (ending in node 15.2; Figure 8) are shown in Figures 7 and 8, respectively. GO terms enriched significantly in female pathway (ending in node 5.4) indicating significant increase in binding of sperm to zona pellucida (GO:0007339), egg coat formation (GO:0035803), prevention of polyspermy (GO: 0060468), and positive regulation of acrosome reaction (GO: 2000344) (Supplementary Table S9). GO terms enriched along the male pathway (ending in node 15.2) were related to ribosomal function, transcription, translation, DNA replication and repair, protein folding, transportation and catabolism, and cell cycle (Supplementary Table S9).

Transcripts of germline-specific genes were enriched in germ cells

Current sea lamprey reference genome contains one germline-specific chromosome (Chromosome 81; Timoshevskaya et al., 2023). Of the 13 genes located on chromosome 81with

detected expression in our PIP-seq experiment, 11 showed enrichment in germ cell clusters (Figure 9). For example, *hykk (LOC116957691)*, *hykk.5 (LOC116957686)* and *hykk.6 (LOC116957689)* were enriched in both male and female germ cells, while *hykk.4 (LOC116957684)* was only enriched in oogonia (Figure 9). Other chromosome 81 genes such as *spopl (LOC116957694)*, *cdh4.3 (LOC116957687)*, *spopl.1 (LOC116957693)*, and *spopl.2 (LOC116957695)*, were also enriched in germ cells of both sexes (Figure 9).

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A study focused on programmed DNA loss in female lamprey germline identified several germline-specific genes with oocyte-enriched expression such as *birc3*, *lrrc8*, *efna*, and *rbm46* (Timoshevskiy et al., 2025). Our PIP-seq data did not showed enrichment of these transcripts (Supplementary Table S4). A bulk RNA-seq study of gonads from larval chestnut (*Ichthyomyzon castaneus*) and northern brook lamprey (*I. fossor*) showed *zp2*, *zp4*, and *nxph1* expression in differentiated female gonads (Ajmani et al., 2021). We found the *zp* genes enriched in oocytes/oogonia in our PIP-seq results (Supplementary Table S4). Furthermore, we found *pfn2* and *mmp19* enriched in immune cells, and *zp* and *alox5* enriched in oogonia and oocytes (Supplementary Figure S18). Our PIP-seq results are consistent with bulk RNA-seq results, and further distinguished genes expressed in germ cells vs. somatic cells.

Sex determination and sexual differentiation in sea lamprey may be in an evolutionary transition state

Sex has a genetic basis among the lower vertebrates despite the frequent absence of recognizable sex chromosomes (Yarnamoto, 1953; Richards & Nace, 1978; Hunter et al., 1983; Yamazki, 1983; Nagahama et al., 2021). The underlying sex regulatory genes appear to have

converged on the doublesex and male abnormal-3 (*mab-3*)-related transcription factor *dmrt* gene family in many species (Kopp, 2012; Zarkower, 2001).

Although the mechanisms underlying germ cell fate specification are diverse in animals, several properties of PGCs appear to be conserved. One property is that in PGCs the somatic differentiation program was repressed. In *Caenorhabditis elegans* and *Drosophila melanogaster*, this is achieved through transient global repression of the RNA polymerase II (RNApII) activity by Pie-1 and the polar granule component (Nakamura et al., 1996; Seydoux et al., 1996). Pie-1 inhibits transcription initiation and elongation activities of RNApII by distinct mechanisms (Ghosh & Seydoux, 2008). Pie-1 inhibits the positive transcriptional elongation factor b (P-Tefb), which promotes the transcription elongation activity of RNAPII by phosphorylating the Ser-2 of the carboxy-terminal domain (CTD) of RNApII (Zhang et al., 2003). The polar granule component inhibits the recruitment of p-Tefb to transcription sites, therefore blocking the transcription initiation (Hanyu-Nakamura et al., 2008). In contrast, in mice, germ cell specification requires active transcription, and the transcription repressor *BLIMP1* specifically shuts off gene expression for the somatic mesodermal program (Kurimoto et al., 2008). In sea lamprey, *srcap* transcript (*LOC116950250*), with 67% identity with *pie-1* mRNA, was enriched in germ cells. However, *blimp1/prdm1* transcripts (*LOC116952900* and *LOC116952967*) were also enriched in germ cells. These results suggest that the mechanism of PGC development in sea lamprey may share characteristics of both invertebrate and vertebrate modes of germ cell specification, potentially representing an intermediate evolutionary state.

Conclusion

Sea lamprey sex determination and sexual differentiation likely involve many genes that interact with environmental factors, representing a transitional state between environmental sex determination and genetic sex determination. This study provides a resource for future studies of germ cell biology and may inform population control effort in sea lamprey**,** a notorious invasive species in the Great Lakes.

Acknowledgements

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Figures:

Figure 1. Cell clusters identified using the “scatter” package in the batch corrected TSNE expression space.

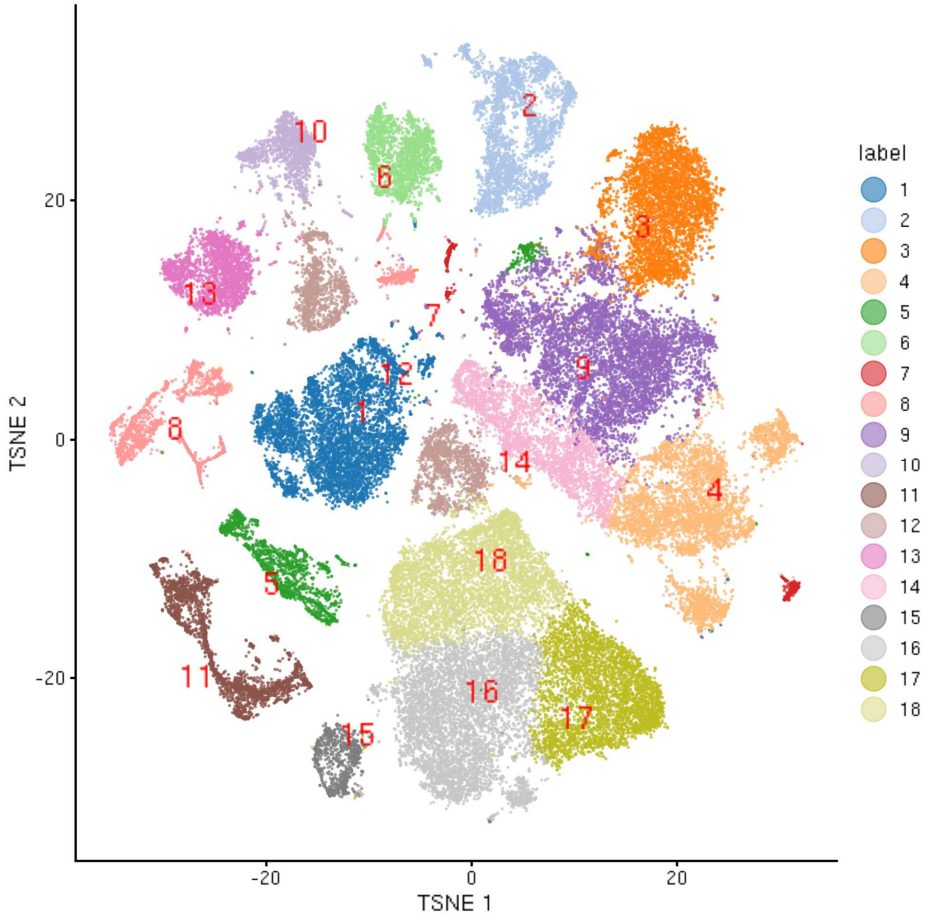


Figure 2. PIWIL1-immunoreactive (Alexa-594: red) migrating primordial germ cells (PGCs) and PGCs in germ cell cysts in larval sea lamprey gonad. Blue: DAPI nuclear stain; Green: autofluorescence; Magenta: DAPI + PIWIL1; White: DAPI + PIWIL1 + autofluorescence; Orange or Yellow: PIWIL1 + autofluorescence. Normal goat serum (NGS) was used as a negative control for immunostaining.

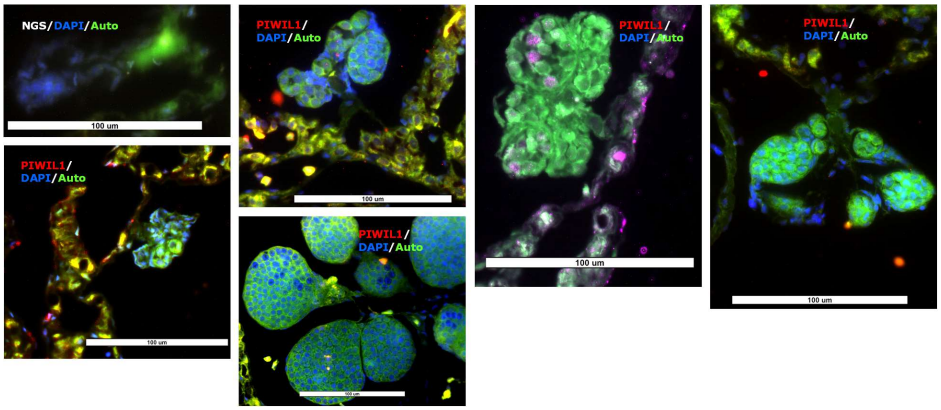


Figure 3. Images of sea lamprey gonadal tissues stain with HCR probes for Piwil1 (yellow) and DAPI nuclear stain (blue). L: larva; NC: negative control without the probes; PM: prespermiating male.

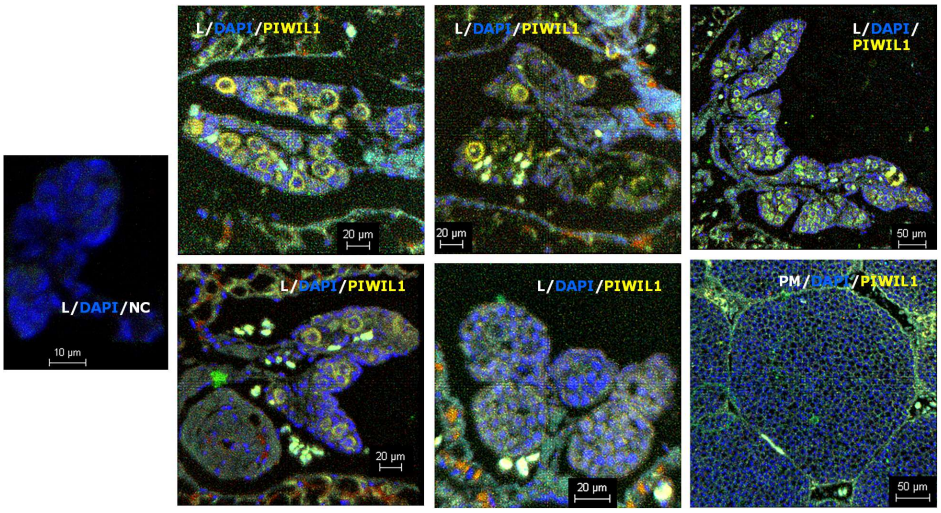


Figure 4. SYCP3-immunoreactive (Alexa 594: red) migrating primordial germ cells and spermatocytes in sea lamprey gonad. Auto: green autofluorescence; Bleu: DAPI nuclear stain; Magenta: DAPI + SYCP3; White: DAPI + SYCP3 + autofluorescence; Orange or Yellow: SYCP3 + autofluorescence. L: larvae; PM: prespermiating male; SM: spermiating male.

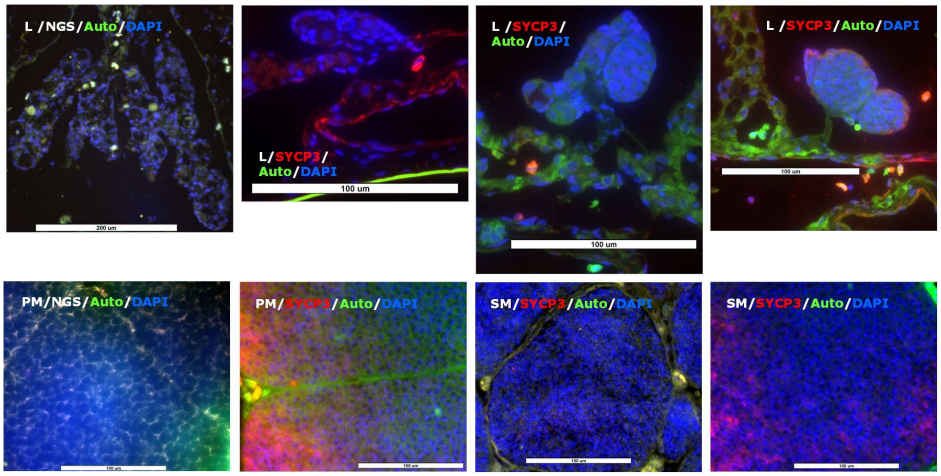


Figure 5. Minimum spanning trees of pseudotime trajectories. Color is the set's mean log expression scaled to its own maximum, so the panels show where each program sits rather than how the programs compare with one another. Size is the percentage of cells in the node detecting the set. Each panel title carries the set's absolute maximum. A panel whose maximum is near zero is showing noise stretched across the full color range.

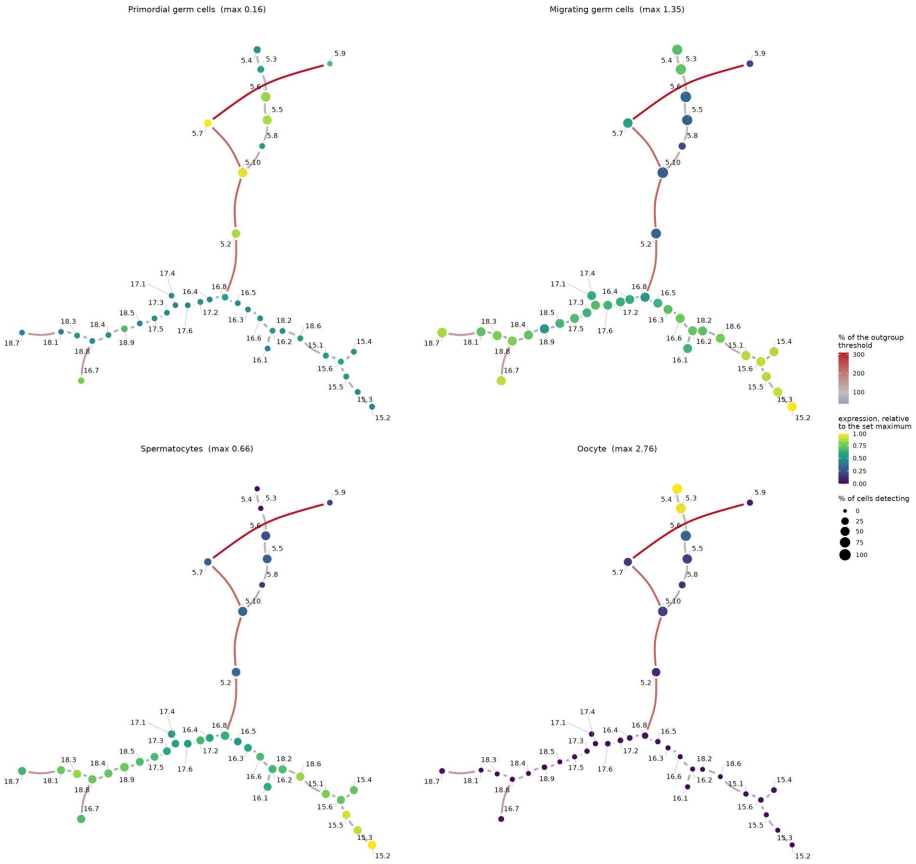


Figure 6. Distribution of the cells and the sample library along the pseudotime.

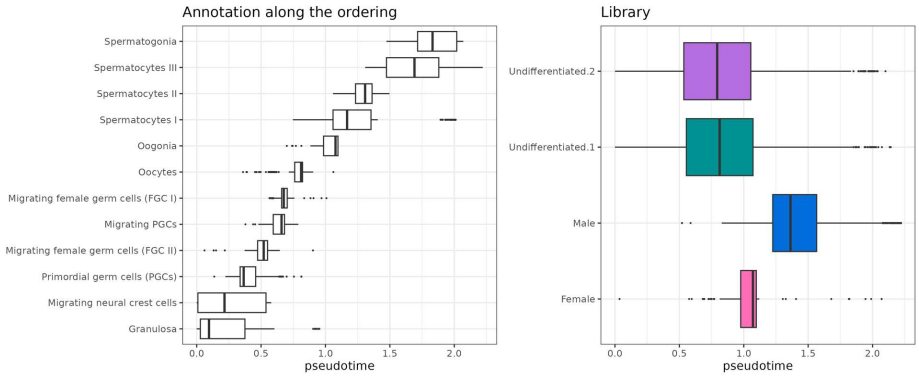


Figure 7. Pseudotime expression heatmaps show the top 25 (by -log10(FDR)) genes significantly up- (left panel) or down- (right panel) regulated along the female trajectory path (ending in node 5.4: oocytes). Genes were tested for pseudotime-associated expression changes using TSCAN::testPseudotime, which fits a natural cubic spline to log-normalized expression along each trajectory path and assesses significance via an F-test (spline vs. null model). P-values were corrected for multiple comparisons using the Benjamini-Hochberg procedure. Columns represent individual cells ordered by TSCAN pseudotime (left to right, early to late). Fill color shows row-scaled (z-scored) log-normalized expression (blue: low; yellow: mid; red: high; limits −4 to 5).

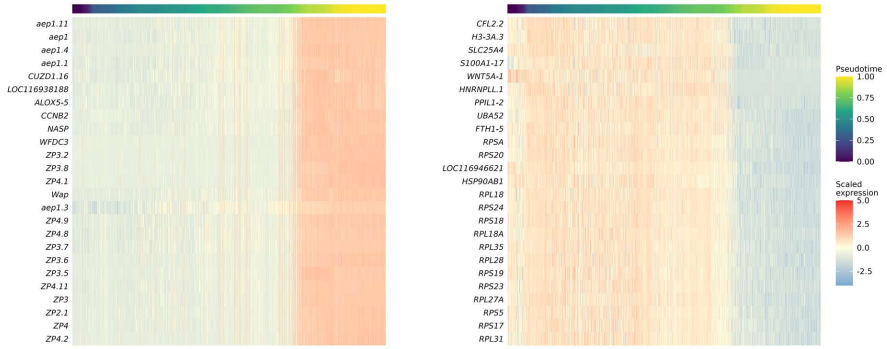


Figure 8. Pseudotime expression heatmaps show the top 25 (by -log10(FDR)) genes significantly up- (left panel) or down- (right panel) regulated along the male trajectory path (ending in node 15.2: spermatocytes). Genes were tested for pseudotime-associated expression changes using TSCAN::testPseudotime, which fits a natural cubic spline to log-normalized expression along each trajectory path and assesses significance via an F-test (spline vs. null model). P-values were corrected for multiple comparisons using the Benjamini-Hochberg procedure. Columns represent individual cells ordered by TSCAN pseudotime (left to right, early to late). Fill color shows row-scaled (z-scored) log-normalized expression (blue: low; yellow: mid; red: high; limits −4 to 5).

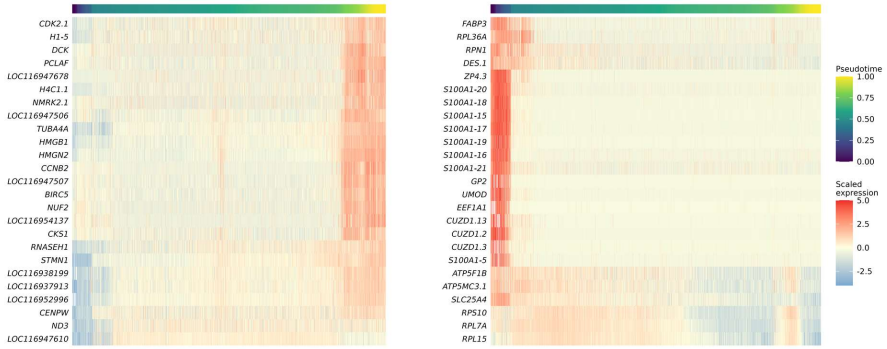


Figure 9. Violin plots show log-normalized expression of Chr. 81 genes across all cell clusters. Internal boxplots show the median and interquartile range. Significance stars indicate FDR-adjusted p-values from pairwise Welch's t-tests comparing each cluster against all others (blocked by sample; genes tested in the upregulated direction; FDR computed under the "some" null hypothesis using findMarkers from the scran package): * FDR < 0.05. Cell counts per cluster are shown in parentheses in the legend.

Commented [YC6]: Need a new figure from Stephanie.